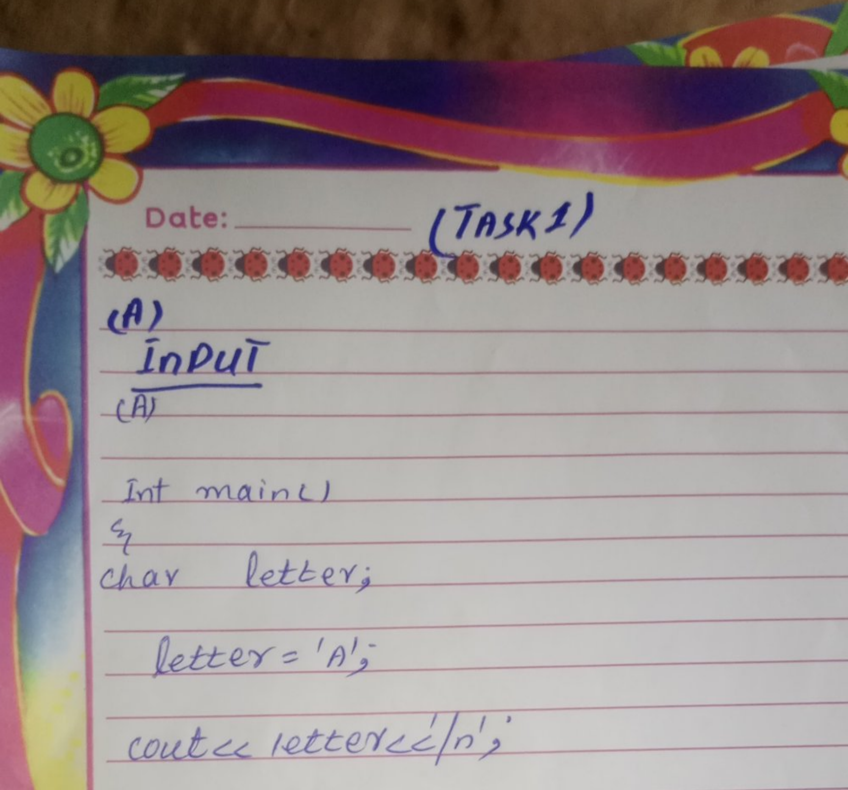
GHAZI UNIVERSITY
DERA GHAZI KHAN

PF LAB

Name	:	M. sharjeel
Program	:	BS Computer Science (BSCS)
Section	:	C
Roll Number	:	12



Date: ________ (TASK 1)

(A)

INPUT

(A)

```
Int main()
{
char letter;
 letter='A';
cout<< letter<<'/n';
```

```
   letter = 'B';

cout<< letter <<'\n';

  cin>> letter;

   returno;
 }
```

OUTPUT

```
A 12142B12142
```

Darsi Notes

Enter

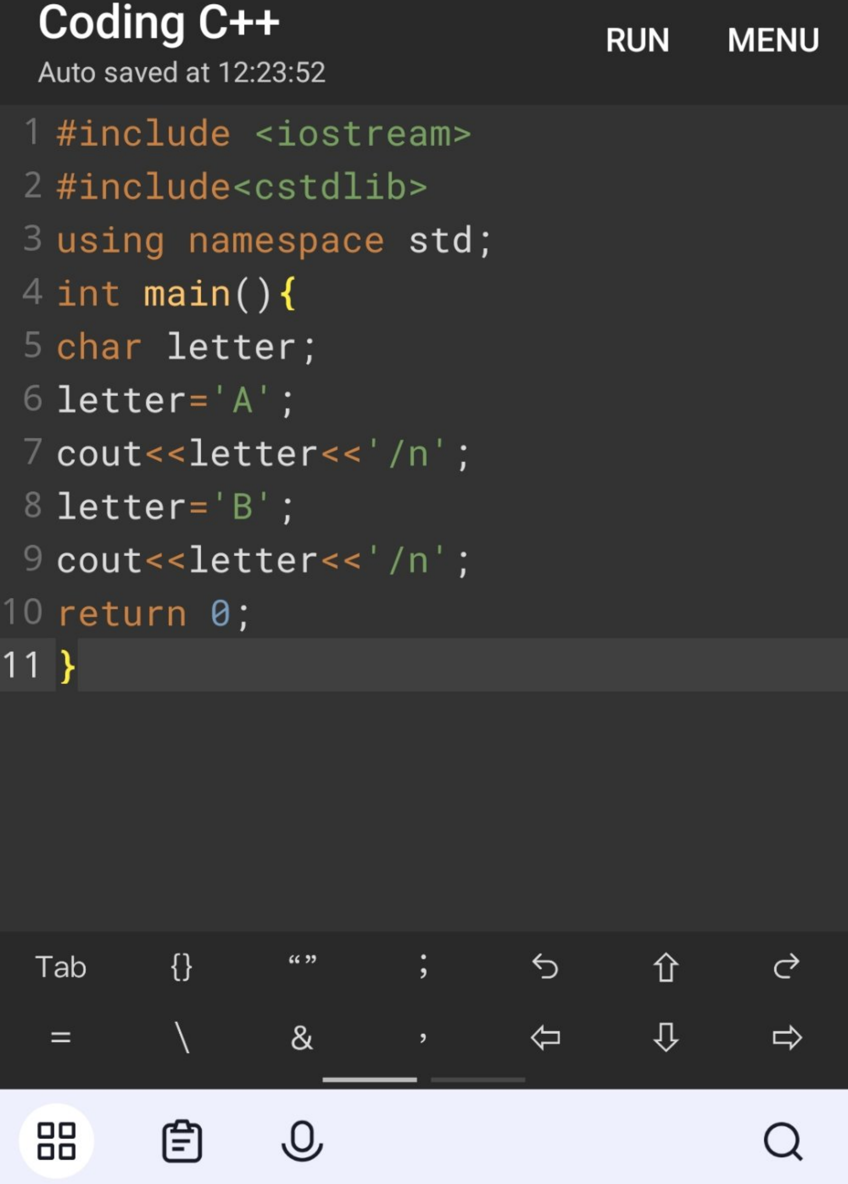
tion =

iv- Division = 2

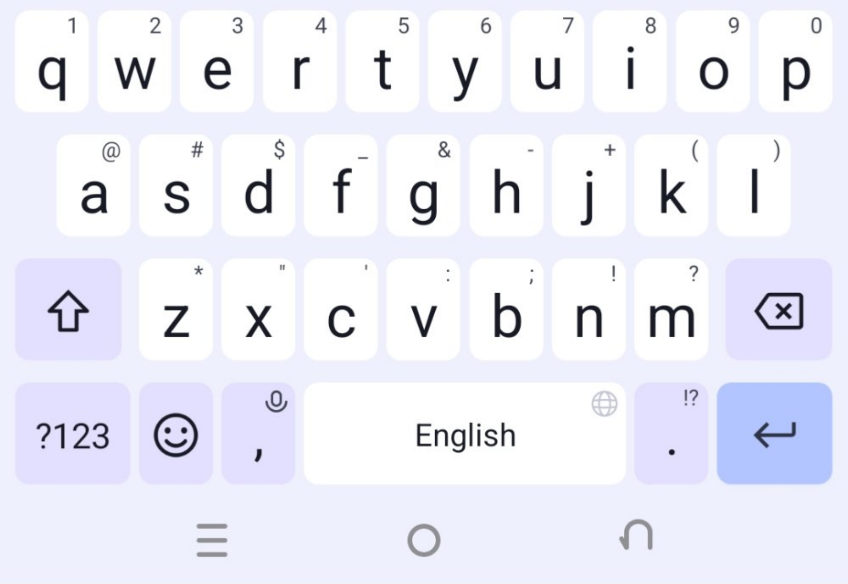
CAMON 20

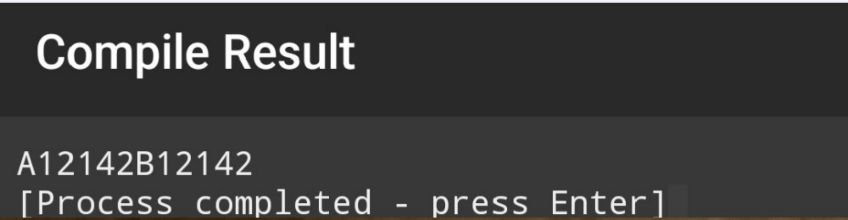
24mm f/1.7 1/50s ISO207

12:24 PM


Coding C++
Auto saved at 12:23:52
RUN
MENU
Tab
{}
“ ”
;
=
\
&
,

```
#include <iostream>
#include<cstdlib>
using namespace std;
int main(){
char letter;
letter='A';
cout<<letter<<'/n';
letter='B';
cout<<letter<<'/n';
return 0;
}
```


q w e r t y u i o p
1 2 3 4 5 6 7 8 9 0
a s d f g h j k l
@ # $ _ & - + ( )
z x c v b n m
* " ' : ; ! ?
?123
English
!?



Compile Result

```
A12142B12142
[Process completed - press Enter]
```

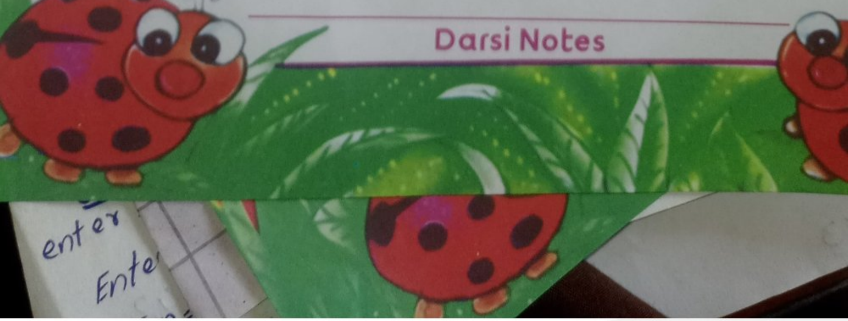
om the
on
Date:

TASK(B)

```
 Int  main( )
{
Char letter;
  letter= 65;
cout<< letter<<'/n';
     letter = 66;
cout<< letter<< '/n';
     cin>> letter;
 return 0;
}
```

OUT PUT

```
A 12142B12142
```


Darsi Notes
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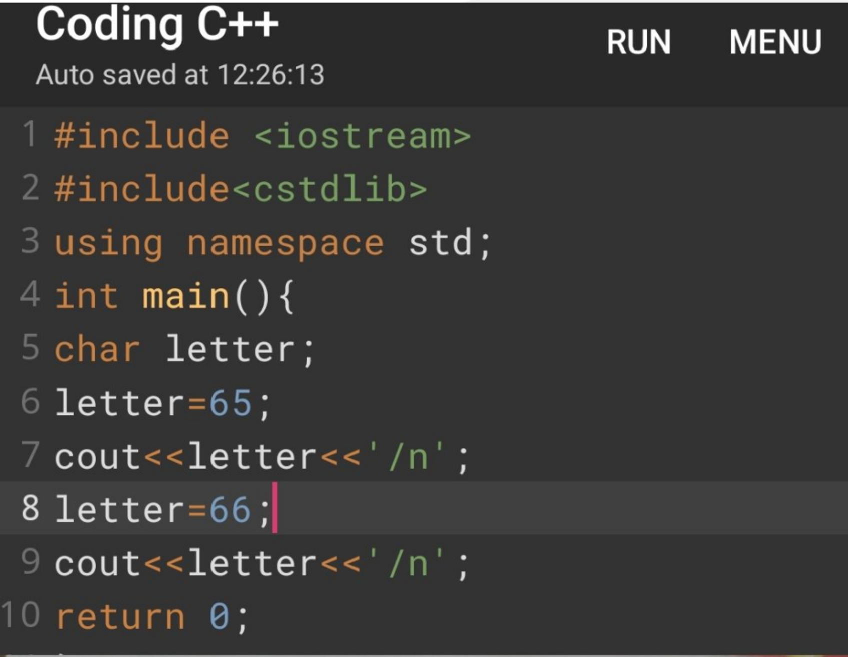
Ente

CAMON 20 •
24mm f/1.7 1/50s ISO207


Coding C++
Auto saved at 12:26:13
RUN
MENU

```cpp
#include <iostream>
#include<cstdlib>
using namespace std;
int main(){
char letter;
letter=65;
cout<<letter<<'/n';
letter=66;
cout<<letter<<'/n';
return 0;
```

(C)

Input

```
Int main()
{
char letter;
   letter='A';
cout<< letter<<'/n';
   letter= 'B';
cout<< letter <<'/n';
      cin>> letter;
   return0;
}
```

OUTPUT

```
A12 142B 12142
```

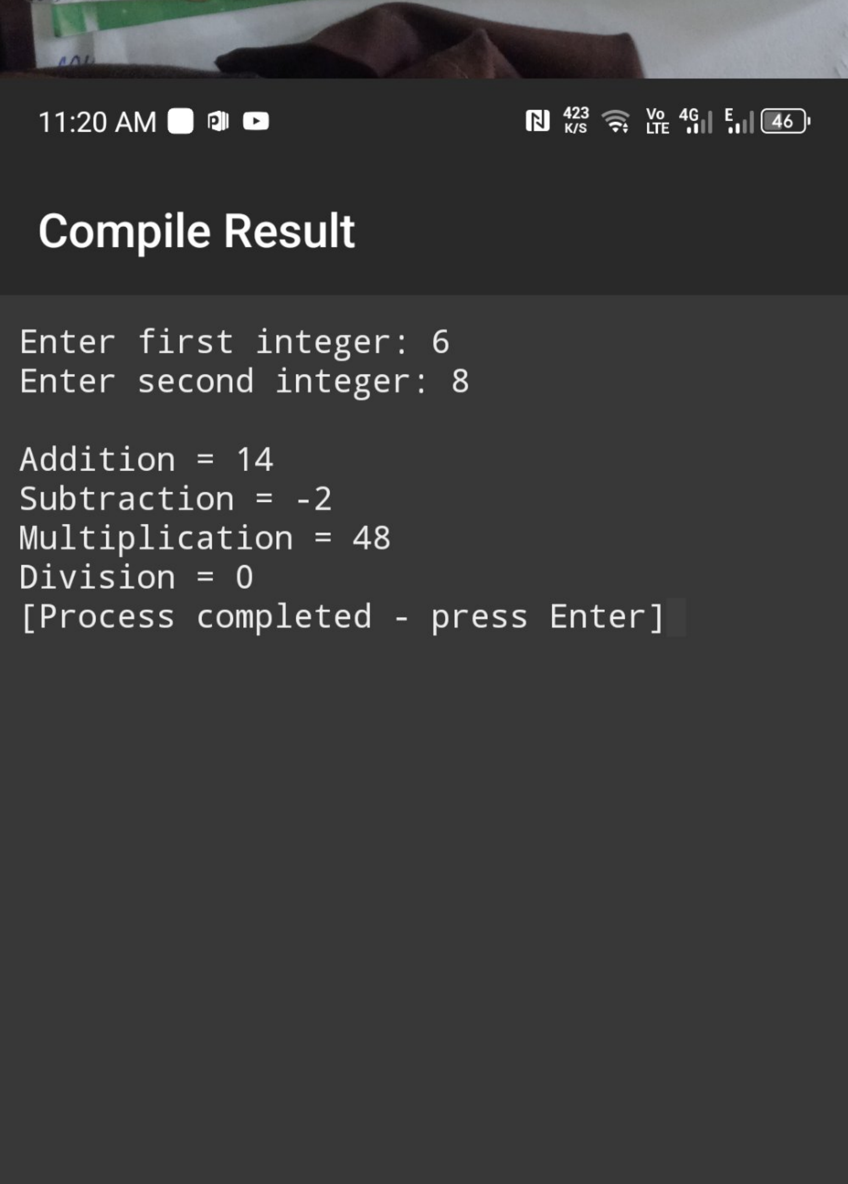
Take two integers as input from the user apply athematic operations on them (+,-,x,/) as print them one secreen.

InpuT

```
Int main()
{
int num1, num2;
print (" Enter Fist integers:")
  Scan F ("%d", & num 1);
print F ("Enter second intengers:")
 scan F ("%d", & num 2);
 print ("/n addition=%d", num1+num2);
print ("/n subtraction = %d", num1-num2);
 print ("/n multiplication=%d",num1*num2);
    IF (num2! =0)
 print ("/n division =%d", num1/num2);
  else
print ( % Division is noT possible by zero.");
      return 0;
```

OuT PuT

```
enter = First integer=6
 Enter second interger=8
i-Adition=4          iii- multiplication=48
ii-subtritioon=2         iv- Division= 2
```


11:20 AM
423 K/S
Vo LTE
4G
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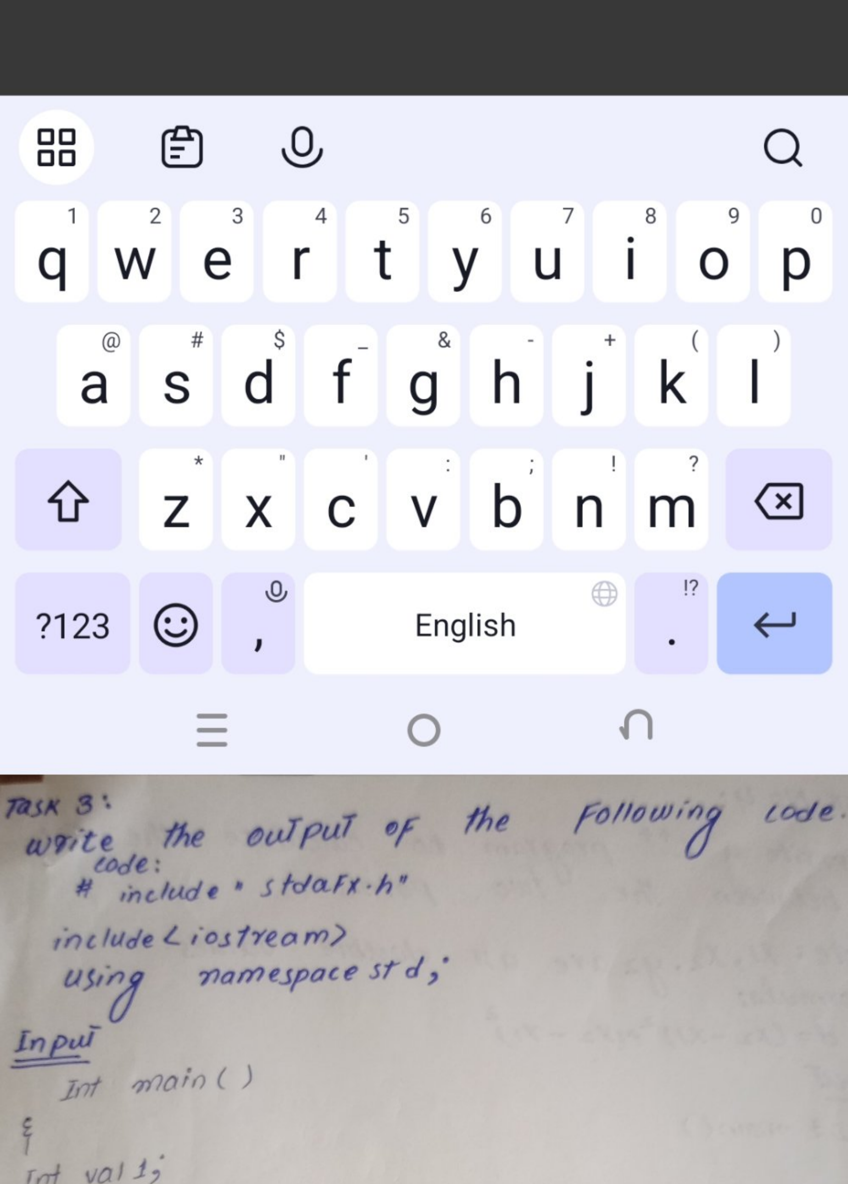
46
Compile Result

```
Enter first integer: 6
Enter second integer: 8

Addition = 14
Subtraction = -2
Multiplication = 48
Division = 0
[Process completed - press Enter]
```

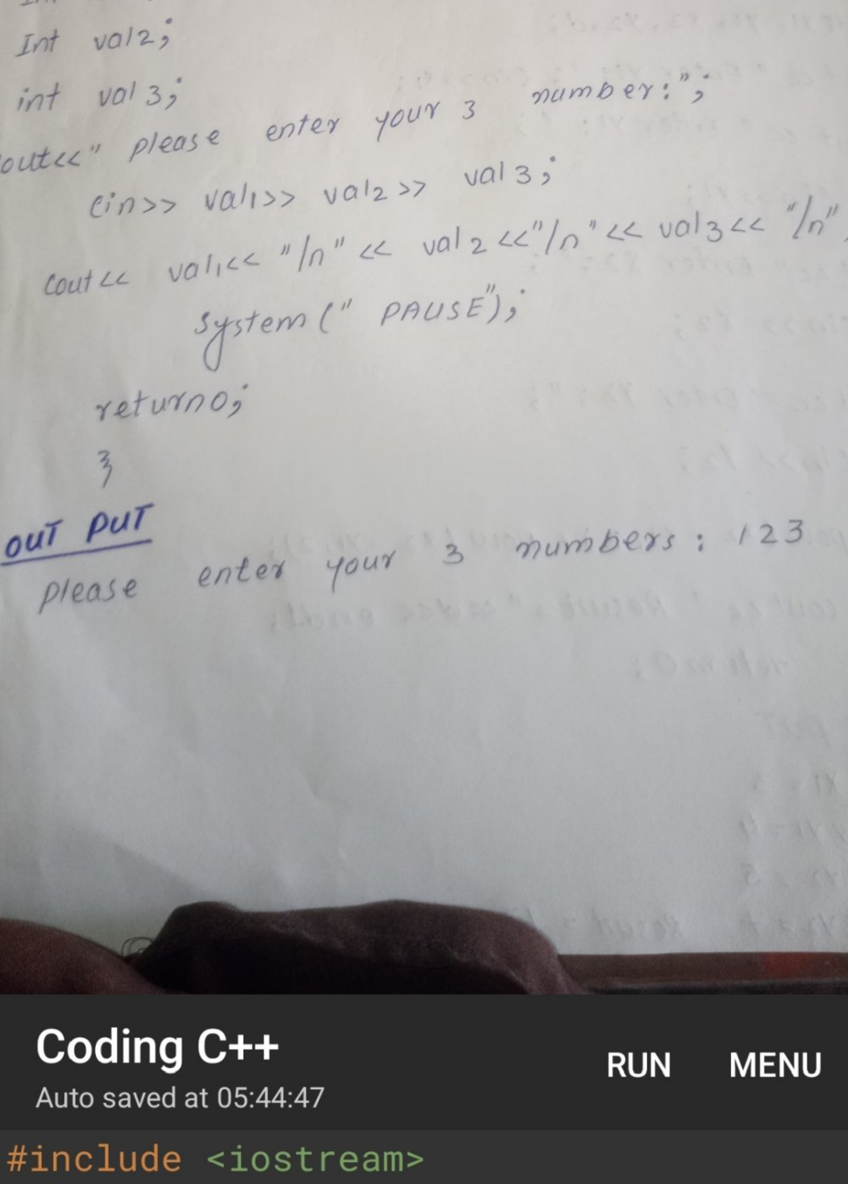

?123
English

TASK 3:
write the output of the following code.
code:

```
# include "stdafx.h"
include<iostream>
using namespace std;
```

Input

```
Int main ( )
{
Int val 1;
```



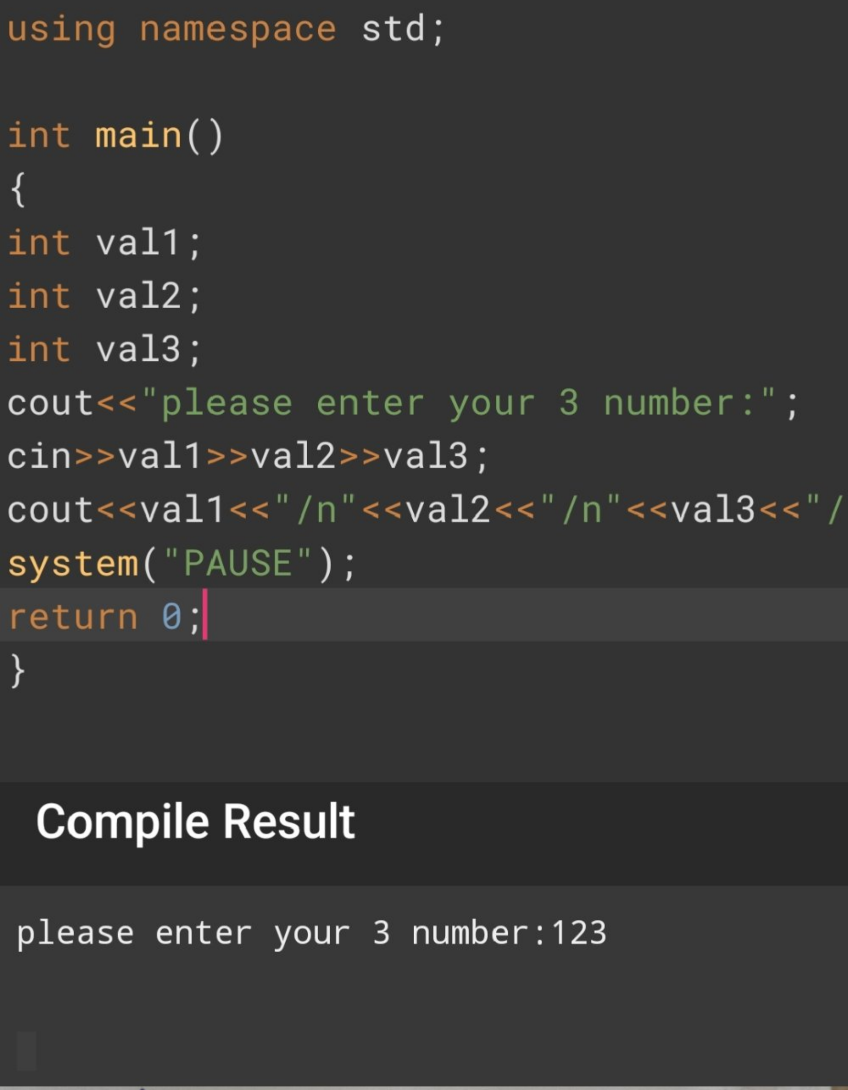
```
Int val2;
int val 3;
out<<" please enter your 3 number:";
   cin>> val1>> val2 >> val 3;
 cout<< val1<< "/n" << val2 <<"/n" << val3<< "/n"
         system(" PAUSE");
   return0;
   }
```

OUT PUT

```
please enter your 3 numbers : 123
```

Coding C++
Auto saved at 05:44:47
RUN
MENU

```
#include <iostream>
```



```
using namespace std;

int main()
{
int val1;
int val2;
int val3;
cout<<"please enter your 3 number:";
cin>>val1>>val2>>val3;
cout<<val1<<"/n"<<val2<<"/n"<<val3<<"/
system("PAUSE");
return 0;
}
```

Compile Result

```
please enter your 3 number:123
```

TASK NO 4:

write a C++ program to calculate the distance between the two points.

Note: x1, x2, y2 are all double values.

Formula:

$$d = (x_2 - x1)^2 + (y_2 - y_1)^2$$

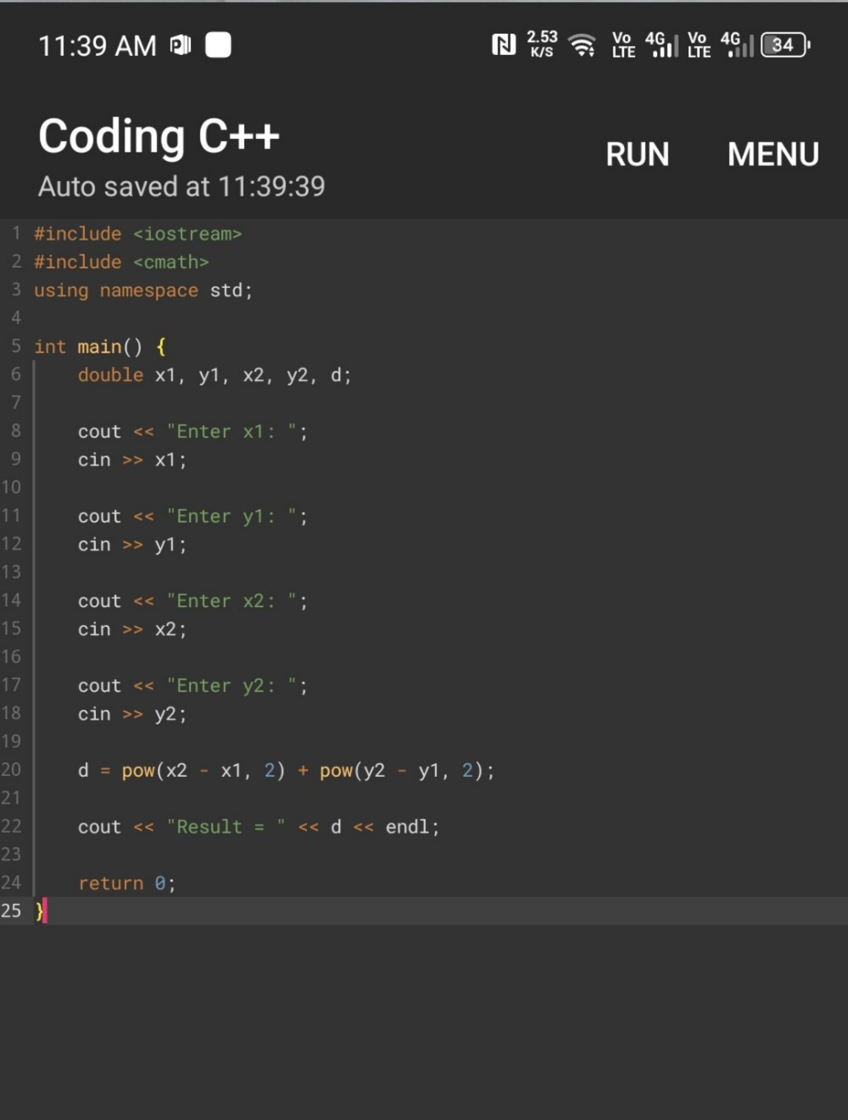
Input

```
  int main()
{
doublex1, y1, x2, y2, d;
  cout << " Enter x1: "; cin >>x1;
  cout << " Enter y1: ";
     cin>> y1;
  cout << " Enter x2:";
     cin >> x2;
  cout << " Enter y2: ";
     cin >> y2;
d = pow (x2 - x1, 2) + pow (y2 - y1, 2);
     cout << " Result= " << d << end1;
       return 0;
```

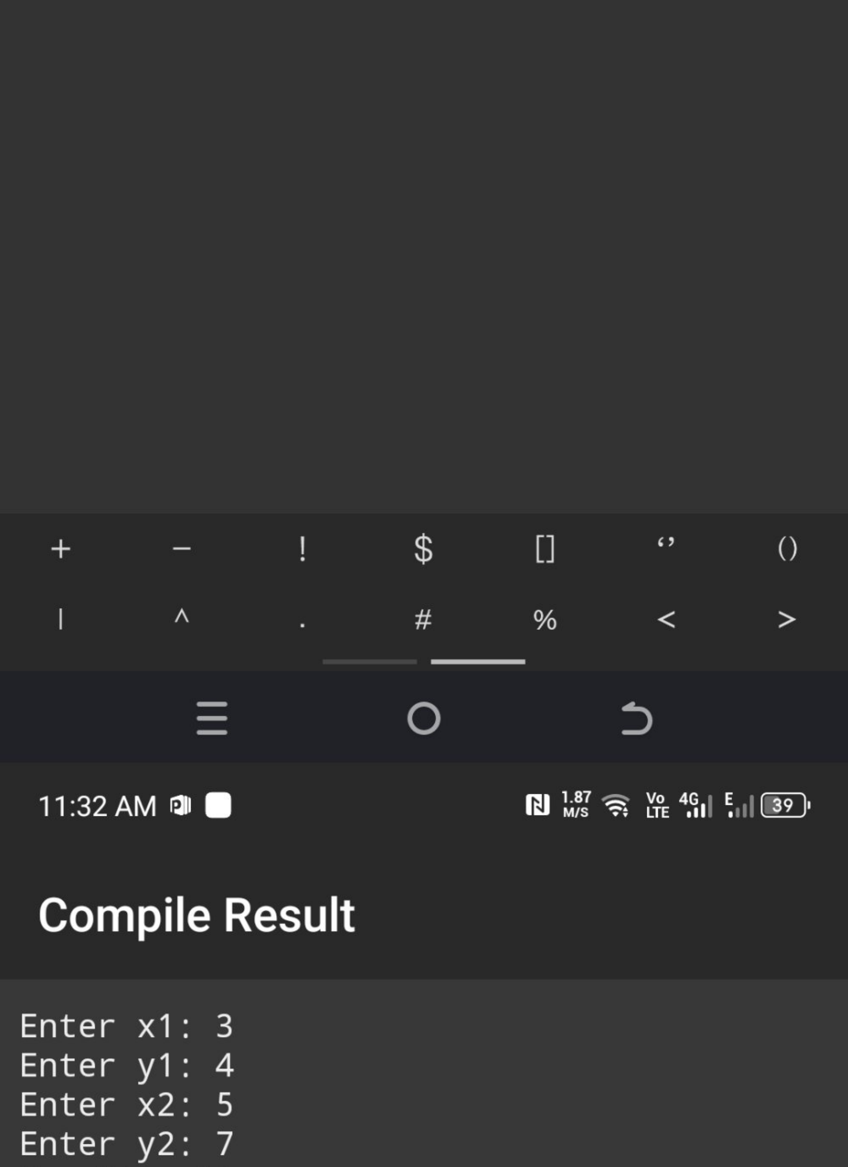
OUT PUT

```
Enter x1 = 3
 Enter y1 = 4
Enter y2 = 5
```

Enter y2 = 7   Result = 13


11:39 AM
2.53 K/S
34
Coding C++
Auto saved at 11:39:39
RUN
MENU

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main() {
    double x1, y1, x2, y2, d;

    cout << "Enter x1: ";
    cin >> x1;

    cout << "Enter y1: ";
    cin >> y1;

    cout << "Enter x2: ";
    cin >> x2;

    cout << "Enter y2: ";
    cin >> y2;

    d = pow(x2 - x1, 2) + pow(y2 - y1, 2);

    cout << "Result = " << d << endl;

    return 0;
}
```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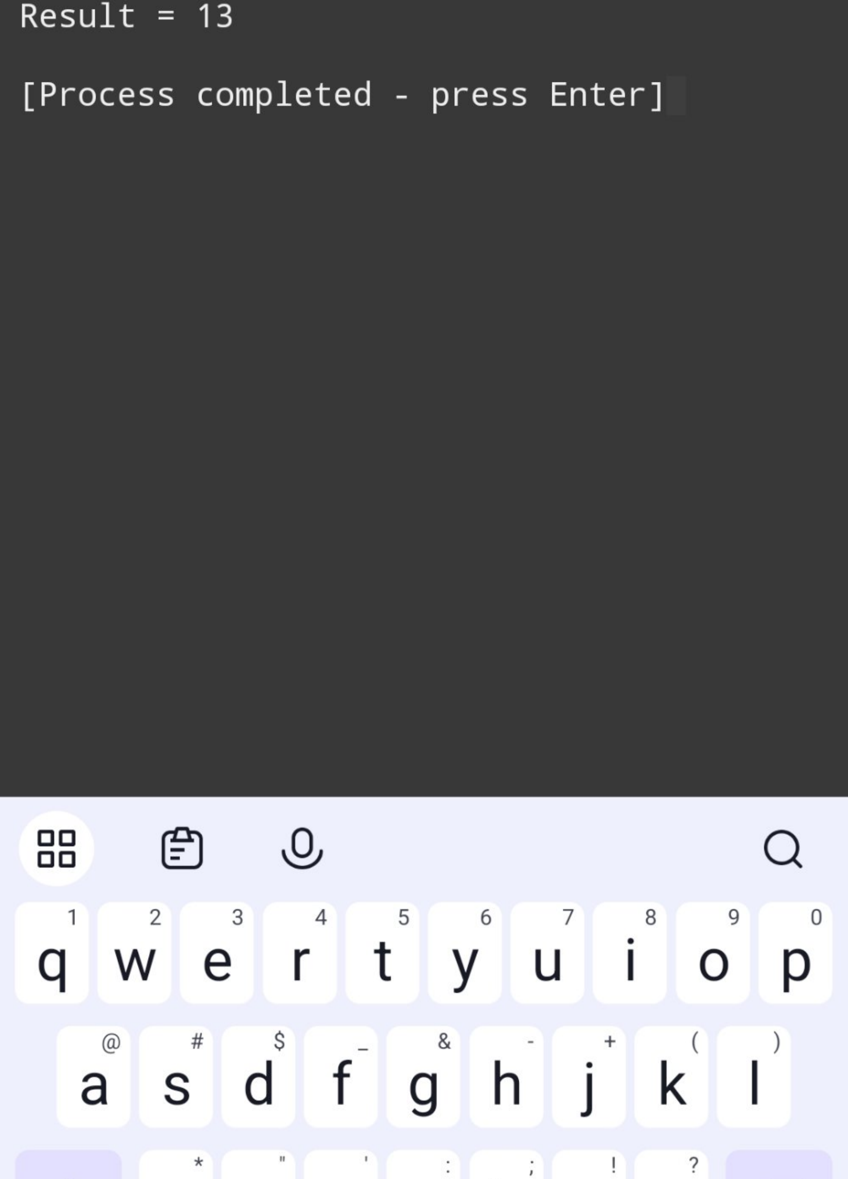
39

Compile Result

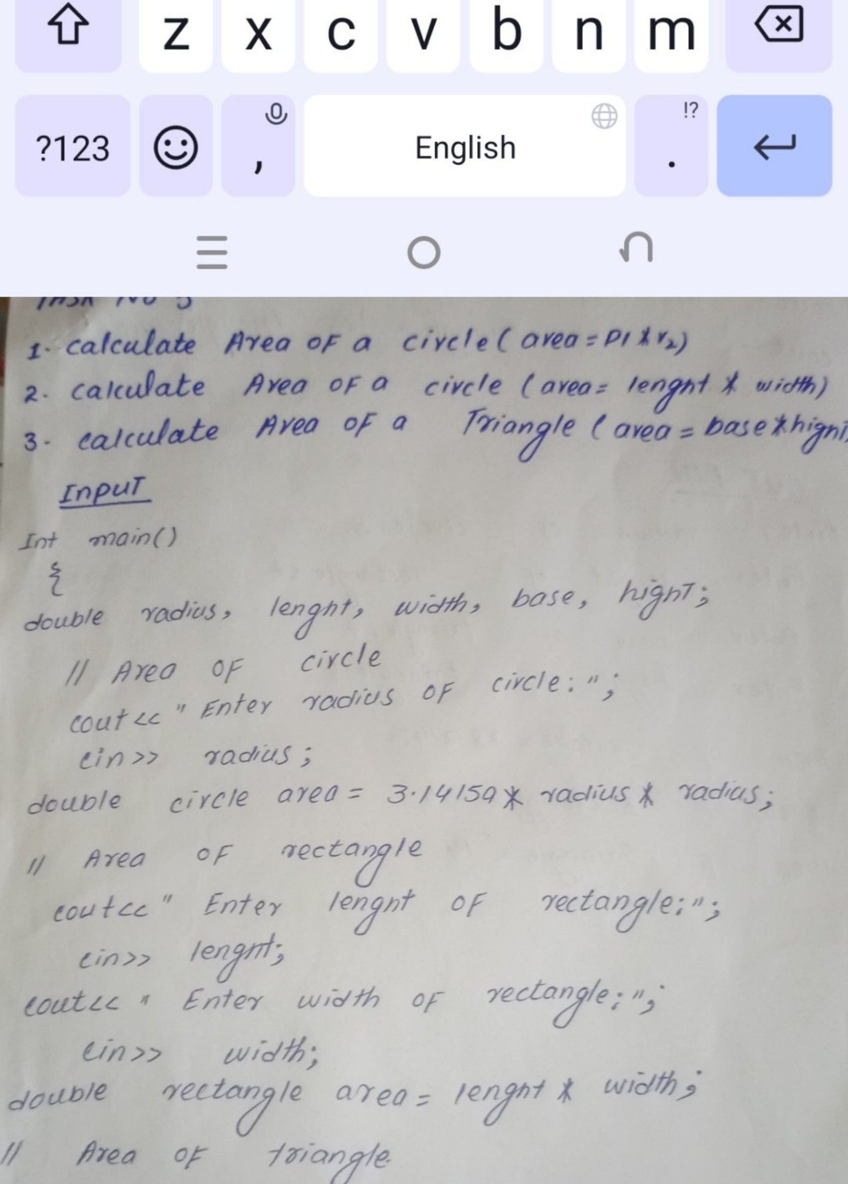
```
Enter x1: 3
Enter y1: 4
Enter x2: 5
Enter y2: 7
```



```
Result = 13

[Process completed - press Enter]
```

1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p
@ # $ _ & - + ( )
a s d f g h j k l
* " ' : ; ! ?


z
x
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b
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m
?123
,
English
.
!?

TASK NO 3

1. calculate Area OF a circle (area = PI * r2)
2. calculate Area OF a circle (area = lenght * width)
3. calculate Area OF a Triangle (area = base * highi

<u>InpuT</u>

```
Int main()
 {
double radius, lenght, width, base, hight;
  // Area OF   circle
  cout<<" Enter radius OF circle:";
  cin>> radius;
double circle area = 3.14159 * radius * radius;
// Area OF rectangle
 cout<<" Enter lenght OF rectangle:";
  cin>> lenght;
cout<<" Enter width OF rectangle:";
  cin>>  width;
double rectangle area = lenght * width;
// Area OF triangle
```

```
cout<< "Enter base of triangle:";
  cin >> base;
out<< "Enter hight of triangle:";
   cin>> hight;
ouble triangle area = 0.5 * base * hight;
/ Display results.
cout<< "\n Area of circle = "<< circle area<< endl;
```

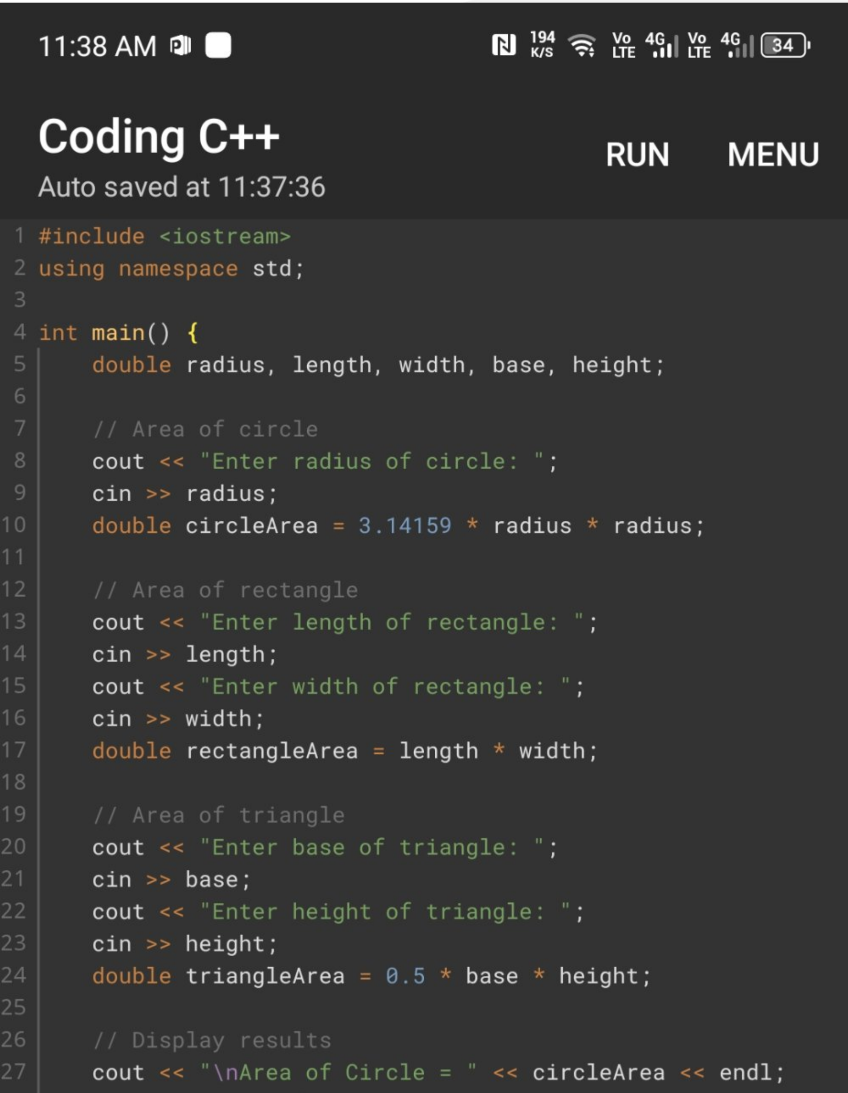
```
cout<<" Area of Rectangle = "<< rectangle Area << en
cout<<" Area of Triangle = "<< triangle Area << en
   return 0;
```

}

OUT PUT

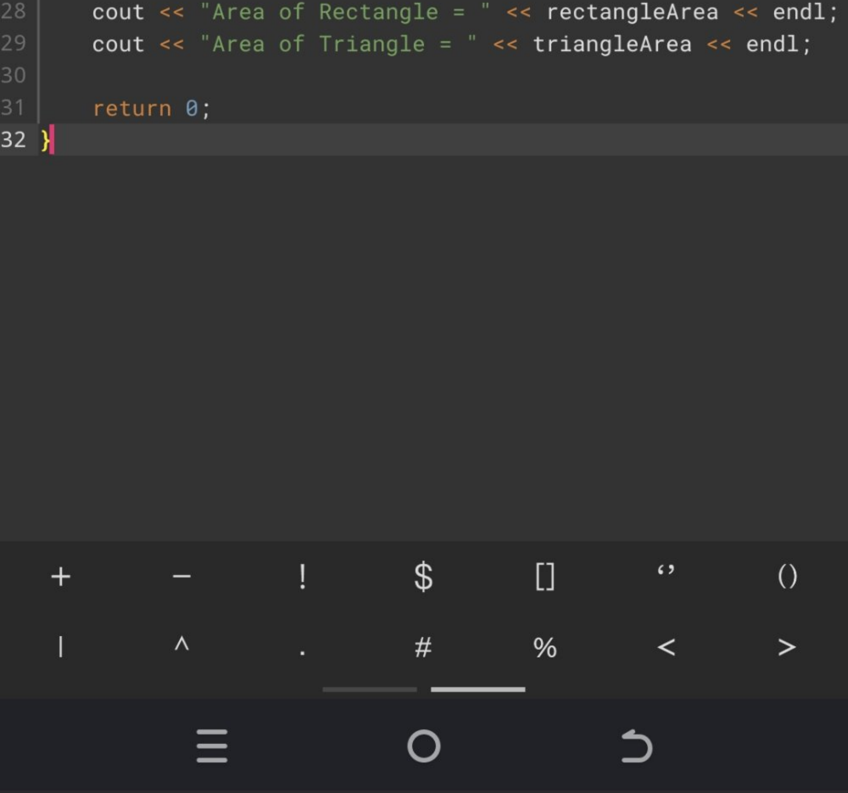
```
Enter radius of circle: 3
Enter lenght of rectangle :6
Enter base of triangle: 7
Enter hight of triangle: 4
Area of circle = 28.2743
Area of Rectangle = 54
Area of Triangle = 14
```

CAMON 20 •
24mm f/1.7 1/50s ISO254

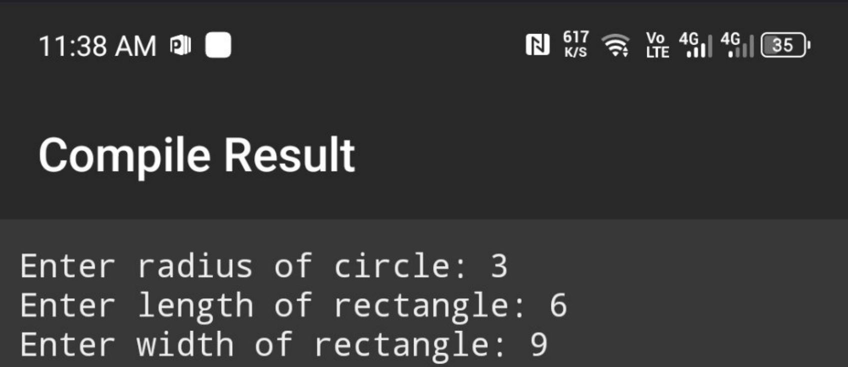


11:38 AM
194 K/S
Vo LTE 4G
Vo LTE 4G
34
Coding C++
Auto saved at 11:37:36
RUN
MENU

```cpp
#include <iostream>
using namespace std;

int main() {
    double radius, length, width, base, height;

    // Area of circle
    cout << "Enter radius of circle: ";
    cin >> radius;
    double circleArea = 3.14159 * radius * radius;

    // Area of rectangle
    cout << "Enter length of rectangle: ";
    cin >> length;
    cout << "Enter width of rectangle: ";
    cin >> width;
    double rectangleArea = length * width;

    // Area of triangle
    cout << "Enter base of triangle: ";
    cin >> base;
    cout << "Enter height of triangle: ";
    cin >> height;
    double triangleArea = 0.5 * base * height;

    // Display results
    cout << "\nArea of Circle = " << circleArea << endl;
```



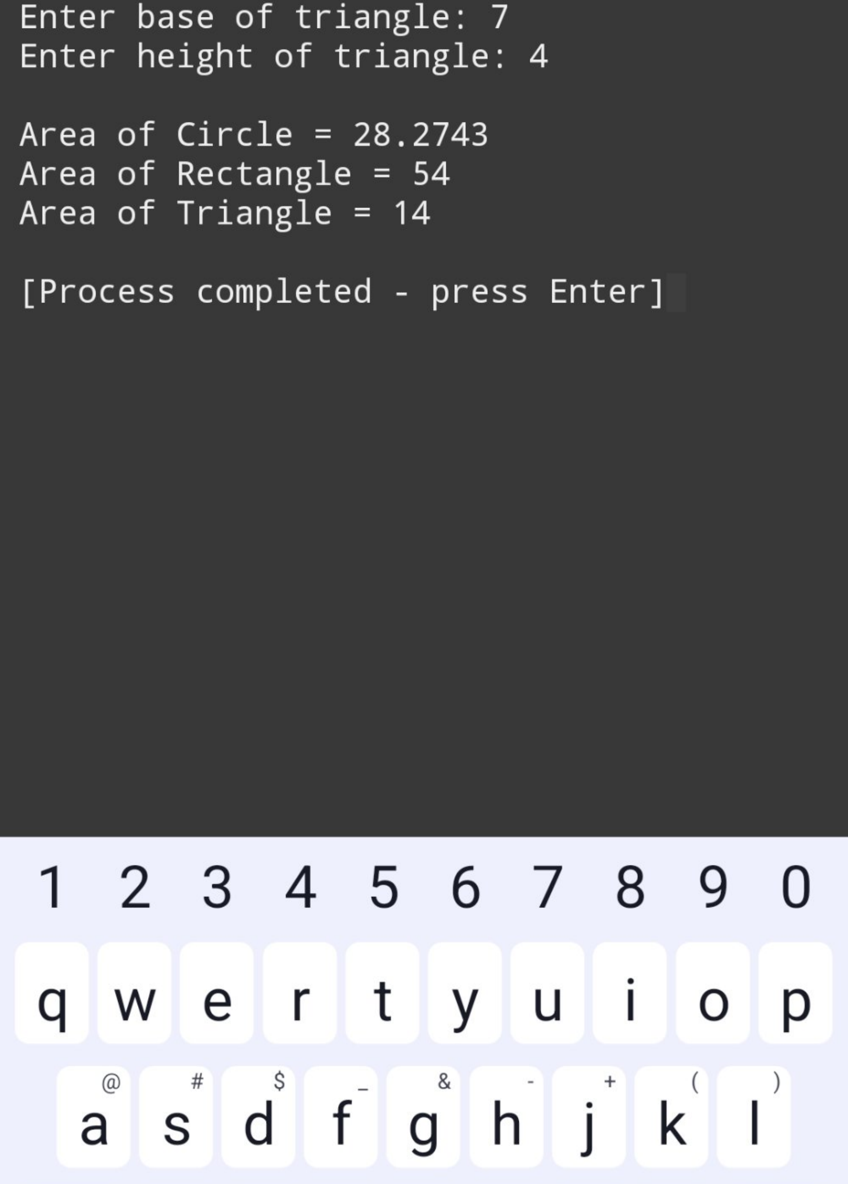
```
    cout << "Area of Rectangle = " << rectangleArea << endl;
    cout << "Area of Triangle = " << triangleArea << endl;

    return 0;
}
```



11:38 AM
617
K/S
Vo
LTE
4G
4G
35

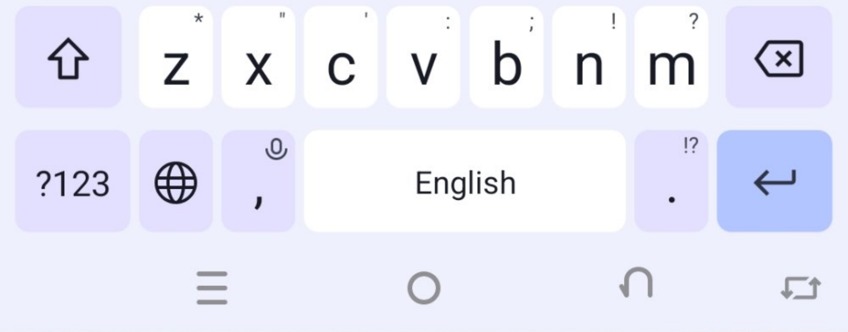
Compile Result

```
Enter radius of circle: 3
Enter length of rectangle: 6
Enter width of rectangle: 9
```



```
Enter base of triangle: 7
Enter height of triangle: 4

Area of Circle = 28.2743
Area of Rectangle = 54
Area of Triangle = 14

[Process completed - press Enter]
```

1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p
@ # $ _ & - + ( )
a s d f g h j k l


z
x
c
v
b
n
m
?123
,
English
.

Task 06

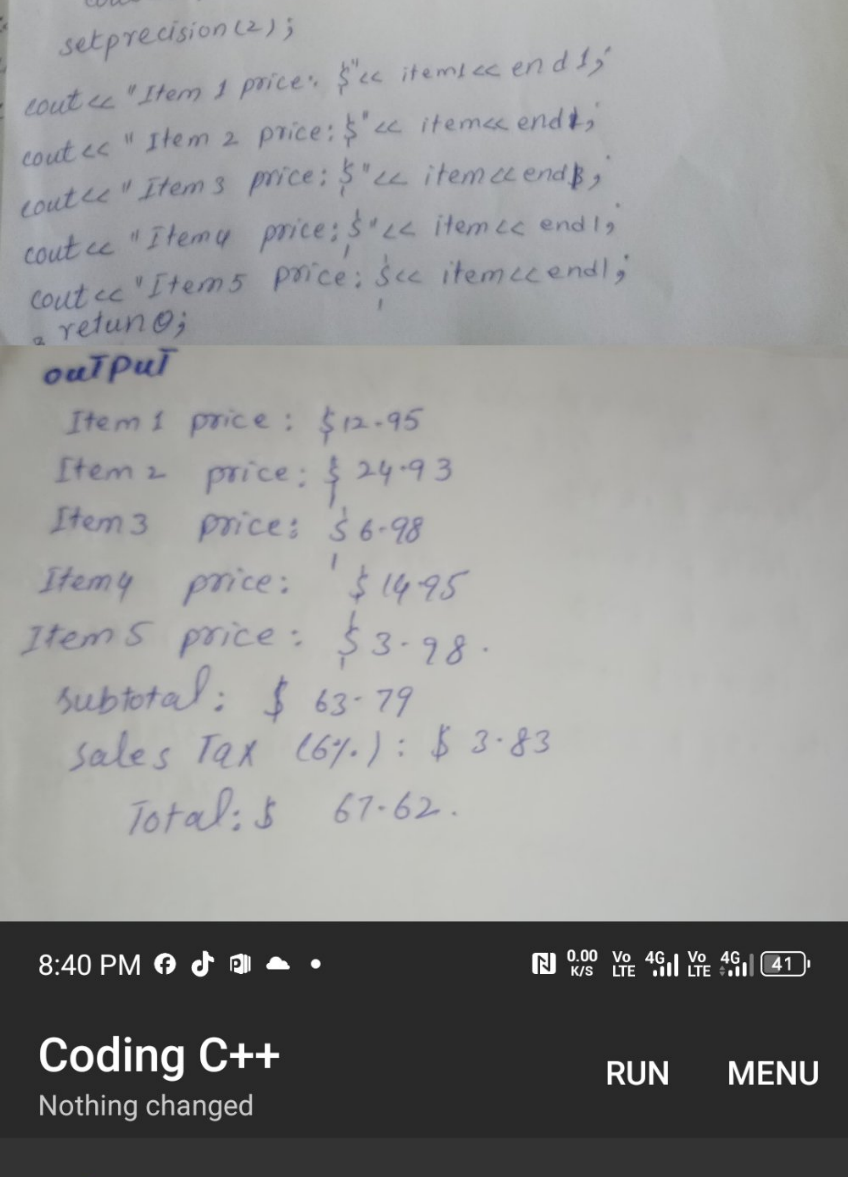
Total purchase A customer in a stronge is purchasing The prices of the Five trem are:

price of item1= $ 12·95
price of item2= $ 24·95
price of item3= $ 6·95
prices of item4= $ 14·95
price of item5= $ 3·95

Write a programe. The prices of the Five items in Five variables. The amount of sales tax, and the total assumethes tax is 6%.

Input

```
main()
{
double item1 = 12·95;
double item2 = 24·93;
double item3 = 6·98;
double item4 = 14·95;
double item5 = 3·98;
double subtotal = item1 + item2 + item3 + item4 + item5;
double sales Tax = subtotal * 0·06;
double total = subtotal + sales Tax;
cout<< Fixed <<
```

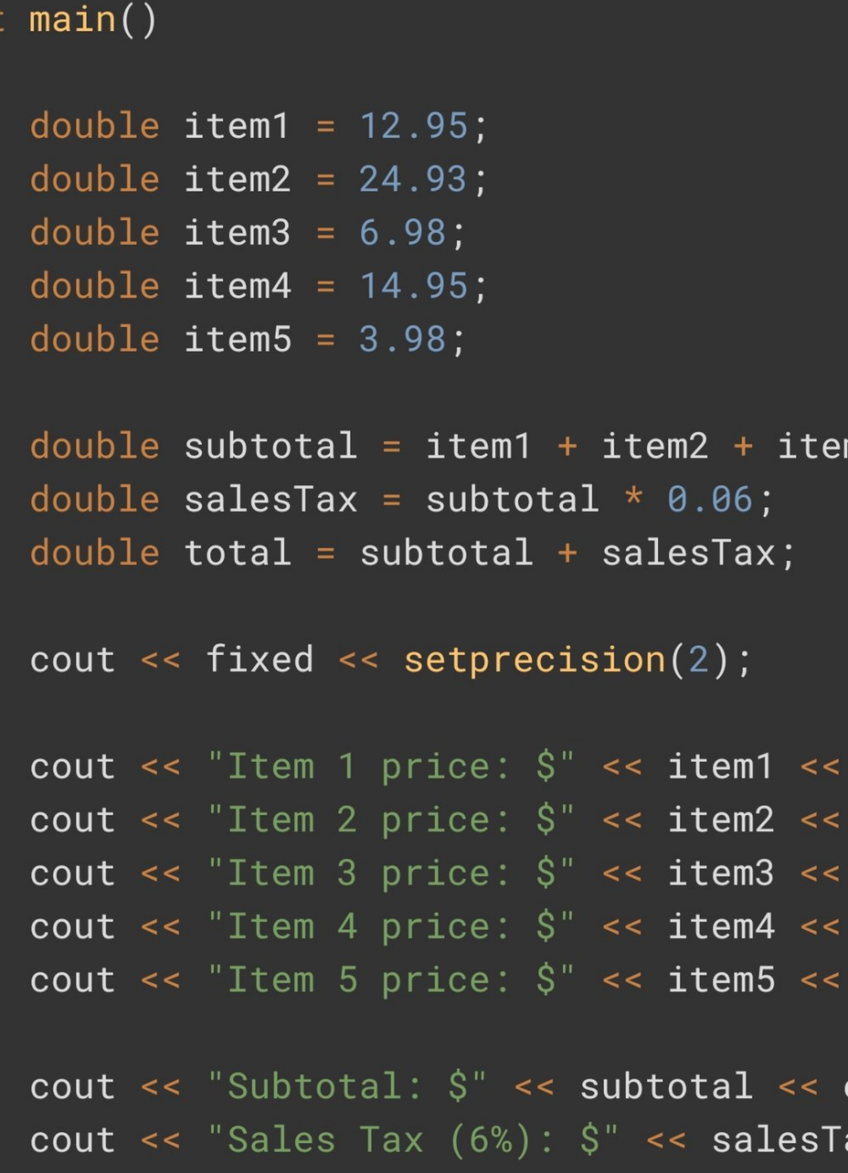


```
setprecision(2);
cout<<"Item 1 price: $"<< item1<< endl;
cout<< " Item 2 price: $"<< item<< endl;
cout<<"Item 3 price: $"<< item<< endl;
cout<< "Item4 price: $"<< item<< endl;
cout<<"Item5 price: $<< item<<endl;
 retun 0;
```

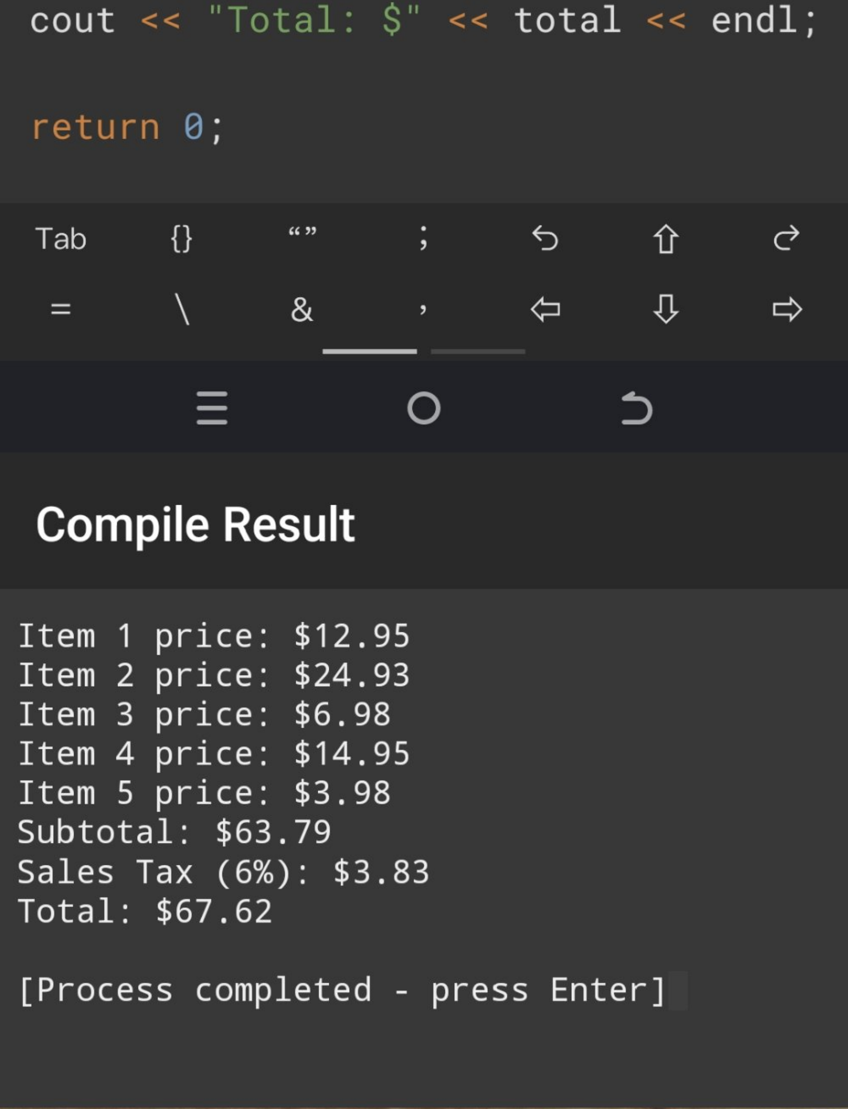
OUTPUT

```
  Item 1 price: $12.95
 Item 2  price: $ 24.93
 Item 3  price: $ 6.98
Item 4  price:   $ 14.95
Item 5 price: $ 3.98.
 subtotal: $ 63.79
  sales Tax (6%): $ 3.83
     Total: $  67.62.
```

8:40 PM
Coding C++
Nothing changed
RUN
MENU



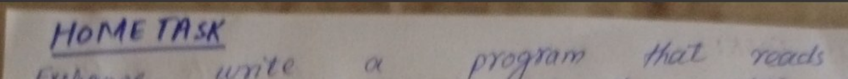
```
main()

double item1 = 12.95;
double item2 = 24.93;
double item3 = 6.98;
double item4 = 14.95;
double item5 = 3.98;

double subtotal = item1 + item2 + ite
double salesTax = subtotal * 0.06;
double total = subtotal + salesTax;

cout << fixed << setprecision(2);

cout << "Item 1 price: $" << item1 <<
cout << "Item 2 price: $" << item2 <<
cout << "Item 3 price: $" << item3 <<
cout << "Item 4 price: $" << item4 <<
cout << "Item 5 price: $" << item5 <<

cout << "Subtotal: $" << subtotal <<
cout << "Sales Tax (6%): $" << salesT
```



```
cout << "Total: $" << total << endl;

return 0;
```

Compile Result

```
Item 1 price: $12.95
Item 2 price: $24.93
Item 3 price: $6.98
Item 4 price: $14.95
Item 5 price: $3.98
Subtotal: $63.79
Sales Tax (6%): $3.83
Total: $67.62

[Process completed - press Enter]
```



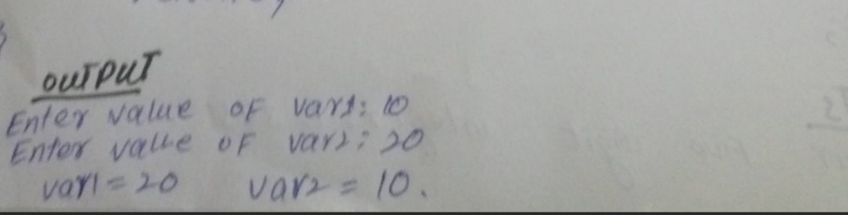
HOME TASK

write a program that reads

Exchange two variables and swaps the value of variables for example if a variable 'va contains 10 and variable 'var2' 20, afte swap operation the Variable 'Var1' con 20 and variable 'var2' contains 10.

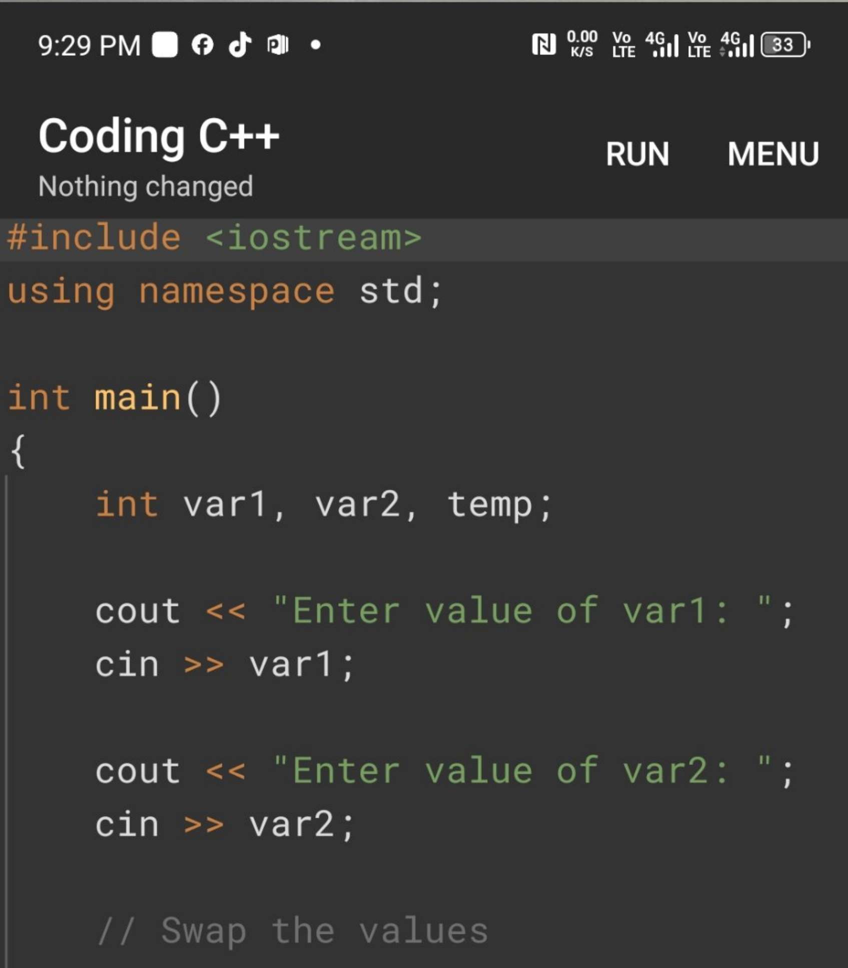
Input

```
# include <Isotream>
 using namespace std;

int main()
{
 int var1, var2, temp;
cout<< " Enter value of var1:";
  cin>> var1;
cout<<" Enter value of var2:";
  cin>> var2;

// swap the values.
  temp= var1;
  var1= var2;
  var2= temp;
cout<<" after swapping:"<< endl;
cout<<"var1 ="<< var1<< endl,
 cout<<" var2 ="<< var2<< endl;
   return 0;
```

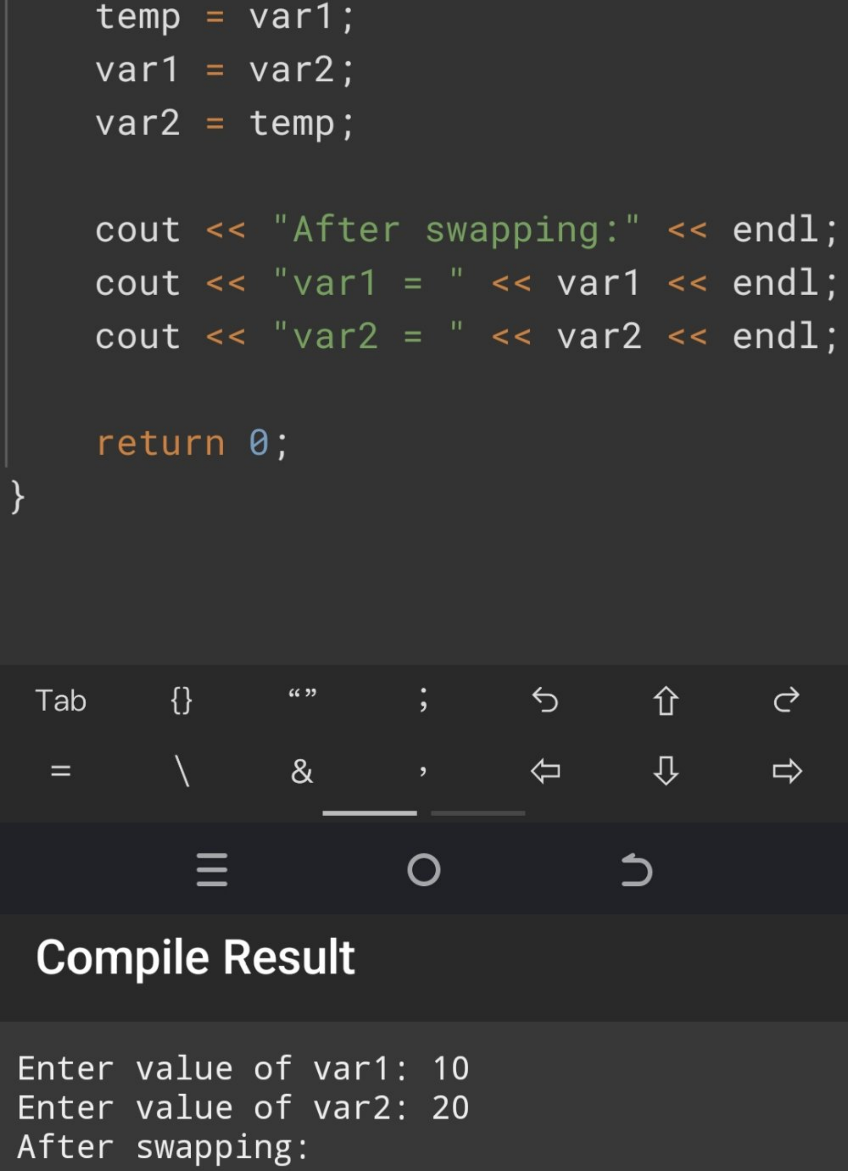


OUTPUT

```
Enter value of var1: 10
Enter value of var2: 20
 var1 = 20     var2 = 10.
```

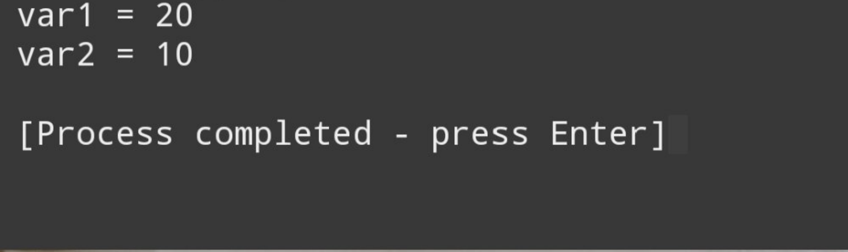

9:29 PM
0.00 K/S
33
Coding C++
Nothing changed
RUN
MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int var1, var2, temp;

    cout << "Enter value of var1: ";
    cin >> var1;

    cout << "Enter value of var2: ";
    cin >> var2;

    // Swap the values
```

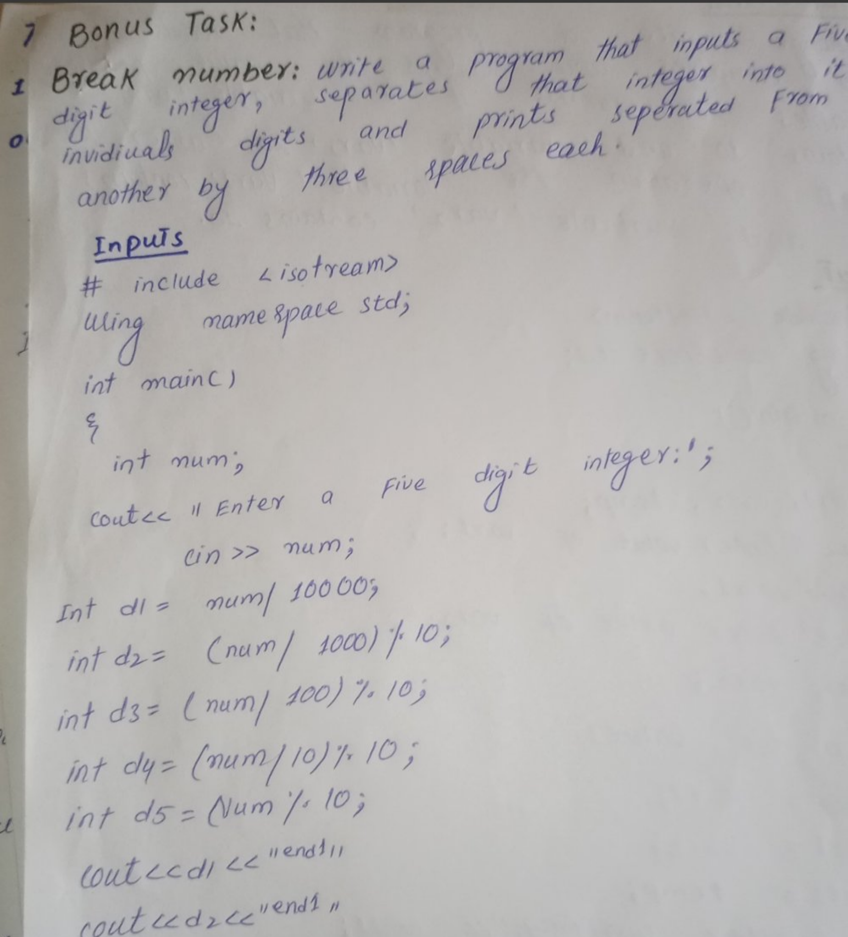

Tab
{}
“”
;
=
\
&
,
Compile Result

```
    temp = var1;
    var1 = var2;
    var2 = temp;

    cout << "After swapping:" << endl;
    cout << "var1 = " << var1 << endl;
    cout << "var2 = " << var2 << endl;

    return 0;
}
```

```
Enter value of var1: 10
Enter value of var2: 20
After swapping:
```



```
var1 = 20
var2 = 10

[Process completed - press Enter]
```

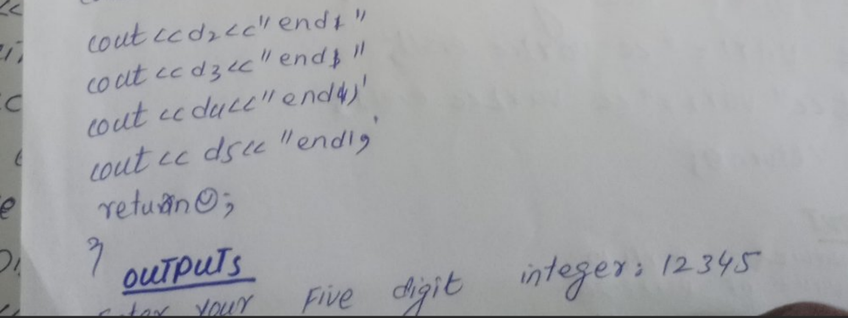


7 Bonus Task:

1 Break number: write a program that inputs a Fiv
digit integer, separates that integer into it
0 invidiuals digits and prints seperated From
another by three spaces each.

Inputs

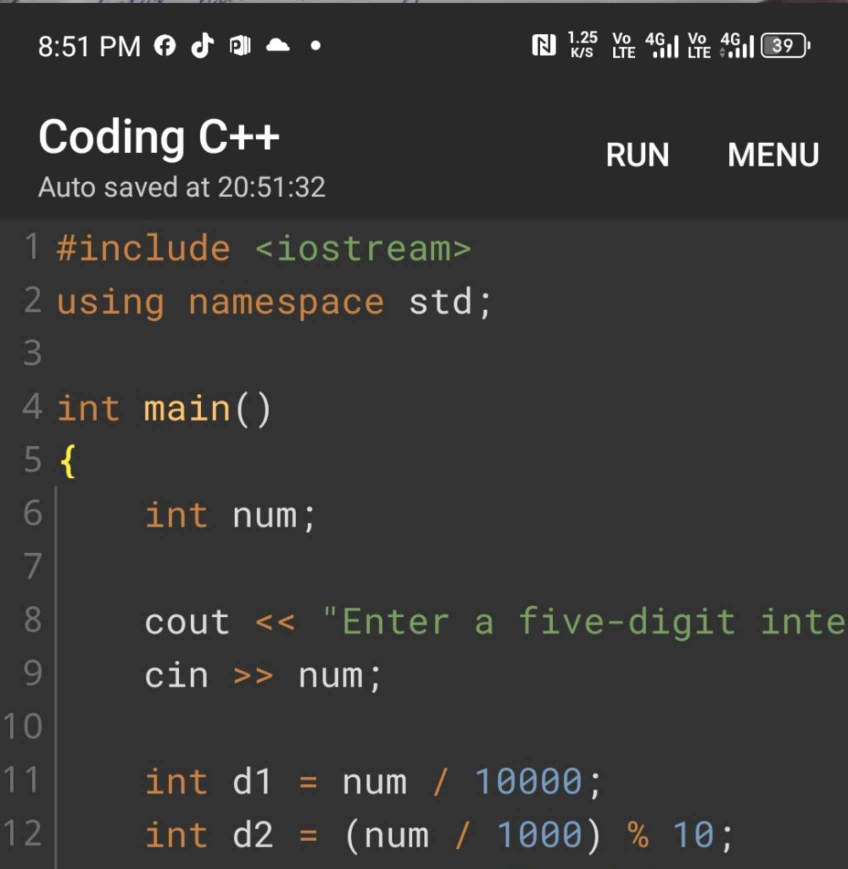
```
# include <isotream>
using namespace std;
int main()
{
  int num;
cout<< " Enter a Five digit integer:';
     cin >> num;
Int d1 = num/ 10000;
int d2= (num/ 1000) % 10;
int d3= (num/ 100) % 10;
int d4= (num/10)% 10;
int d5= Num % 10;
cout<<d1 <<"endl"
cout<<d2<<"endl "
```



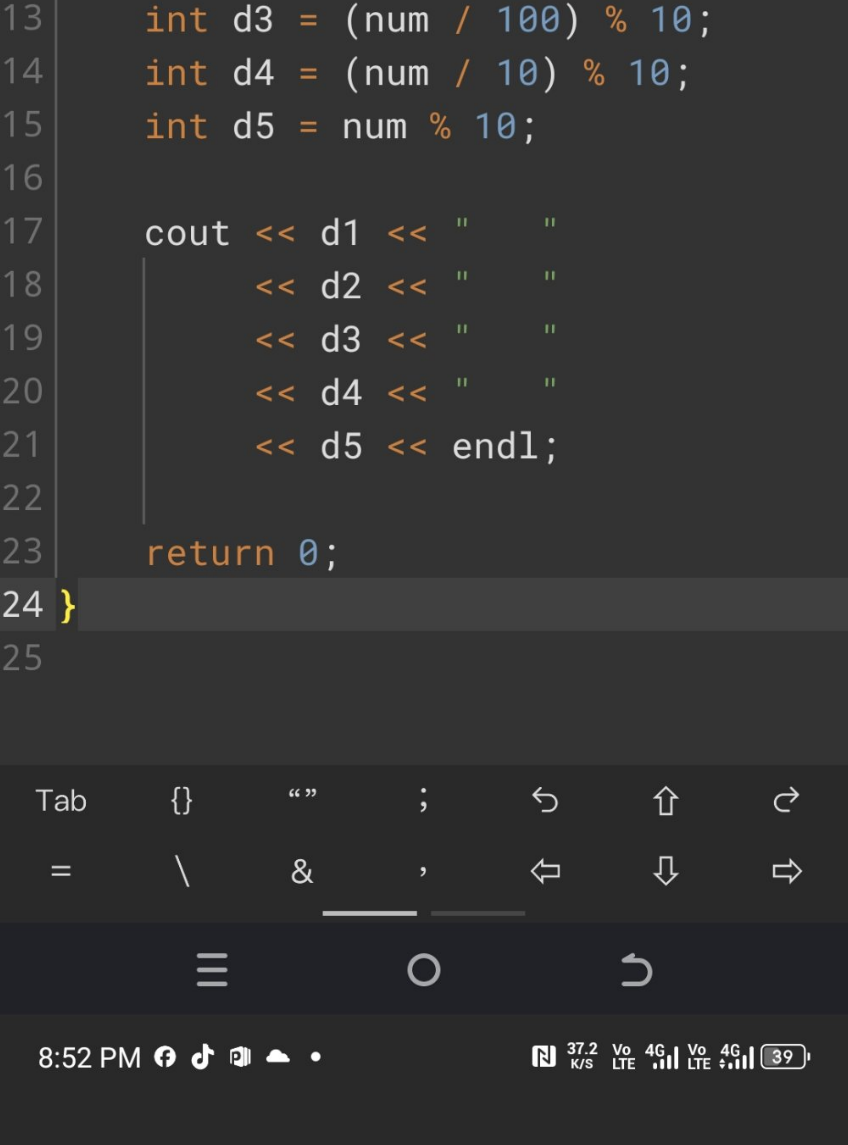
```
cout <<d2<<" endl"
cout <<d3<<" endl"
cout <<d4<<" endl;
cout << d5<< " endl;
return 0;
}
```

OUTPUTS

Enter your Five digit integer: 12345


8:51 PM
Coding C++
Auto saved at 20:51:32
RUN
MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter a five-digit inte
    cin >> num;

    int d1 = num / 10000;
    int d2 = (num / 1000) % 10;
```



```
    int d3 = (num / 100) % 10;
    int d4 = (num / 10) % 10;
    int d5 = num % 10;

    cout << d1 << "   "
         << d2 << "   "
         << d3 << "   "
         << d4 << "   "
         << d5 << endl;

    return 0;
}
```

Tab
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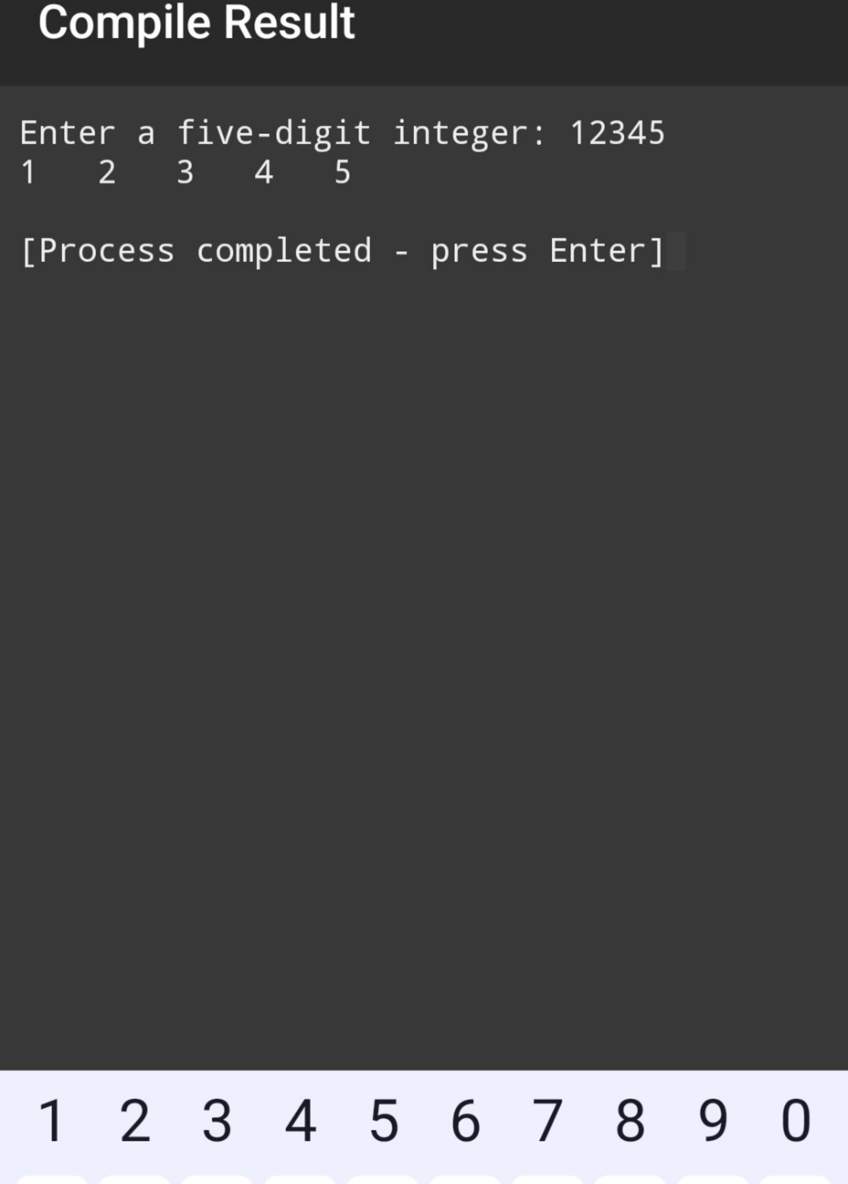
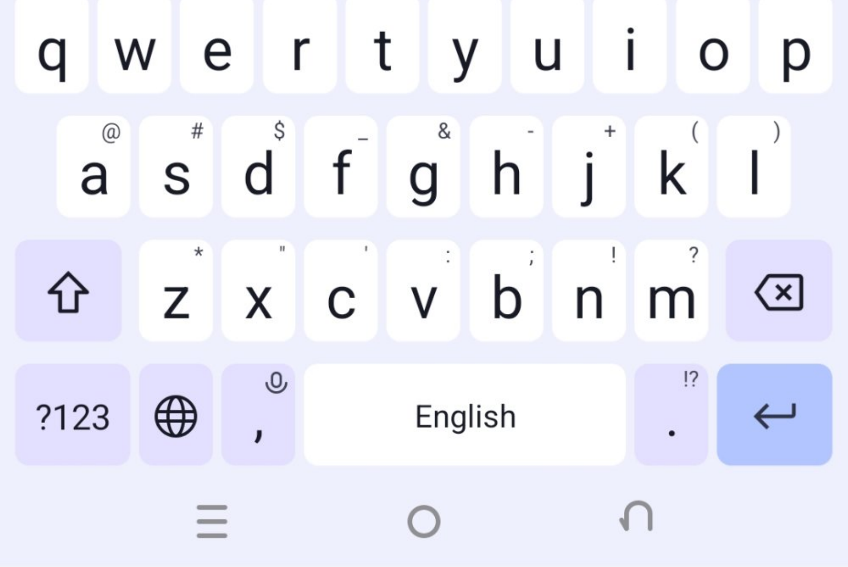

Compile Result
1 2 3 4 5 6 7 8 9 0

```
Enter a five-digit integer: 12345
1   2   3   4   5

[Process completed - press Enter]
```


q w e r t y u i o p
@ # $ _ & - + ( )
a s d f g h j k l
* " ' : ; ! ?
z x c v b n m
?123
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